R6.10: literature closure for rounds 4–5

Date 2026-10-18. **Lead** R6.10 (round 6), branch `research/r6-literature`. **Sources** `RESULTS.md` (one entry per citation, with fetched URLs), `claims_status.json`, `raw/fetch_log.json`, `raw/crossref_hits.jsonl`, `raw/semanticscholar_hits.jsonl`, `references_proposed.md`. No code and no measurements. Every statement below is either checked against a fetched source or marked unverified.

Question and method

Rounds 4–5 cited some results as "recalled", from search snippets only, or with no check. Each citation in the note's §13–§18, in round-4 leads 16.2–16.7, and in R5.2 and R5.4 was checked in two ways:

- **Bibliographic data:** one Crossref record per reference.
- **Content:** ePrint abstract pages and every full text that could be fetched (authors' pages, arXiv, open-access publisher PDFs).

IACR PDF downloads and Springer pages were blocked, so some content checks rest on abstracts or on secondary sources. The table marks which.

Verdict

The table has 47 rows. Bibliographic data was fetched for 46; Vélu 1971 is unverified. By status: 13 confirmed, 4 confirmed (secondary), 14 bib-only (use unverified), 8 corrected, 7 mismatch, 1 unverified, 0 withdrawn.

Four findings change our claims:

1. **4E test (§17.2).** US 7,257,709's cofactor-4 test, $\mathrm{Tr}(x) = 0$, $x \neq 0$, $\mathrm{Tr}(x \cdot H(b/x^2)) = 0$, is already free of square roots and does not use y. Our form's difference is that it needs no inversion. That comparison is derived from the two formulas; no source states it.
2. **Saturation penalty (§15.2, 16.4, R5.1).** Faugère–Huot–Joux–Renault–Vitse (EUROCRYPT 2014, §3.2) give the m^{n-1} loss in decomposition probability for translation-invariant bases, the $\div m$ in relations needed, and say the loss "is more than compensated by the improved resolution of the associated polynomial systems". That anticipates both the μ^{m-2} law (derived from their accounting) and its oracle dependence. They must be cited.
3. **GHS and genus-2 costs (R5.2, §13.4).** The factor g! comes from Gaudry 2000's relation-search term, while $q^{2-2/g}$ is Gaudry–Thomé–Thériault–Diem 2007's exponent. $g! \cdot q^{2-2/g}$ is R5.2's own model. §13.4's "$\tilde{O}(q)$ (Gaudry)" in genus 2 is the GTTD bound; Gaudry 2000 alone gives $\tilde{O}(q^2)$. Enge–Gaudry's $L_q^g[1/2, \sqrt{2}]$ is confirmed from two secondary sources.
4. **GHS02 transport (16.7 §7.4).** According to Maurer–Menezes–Teske 2002, $O(q^{n/4+\varepsilon})$ is the cost of an isogeny between two *given* curves. Finding a weak curve that is isogenous to the target may require a search through the whole weak set.

Smaller corrections:

- The patent dates are confirmed. EGO was published in Adv. Math. Commun. 5 (2011).
- ePrint 2026/749 is a method paper, not a survey.
- The condition "n odd, or m = n, or Tr(a) = 0" is GHS02's theorem, not a "Menezes–Qu condition".
- For prime N, the least magic number is t + 1 with t = ord_N(2) (MMT Thm 11), not "genus $\geq 2^{N-1}$-ish".
- MTW's abstract does not say "fields with many GHS-weak classes".
- Koshelev has a published correction.
- The xedni obstruction is a canonical-height bound on dependency coefficients.
- Kuhn–Struik's $\sqrt{(2rL)}$ (for $L < r^{1/4}$) is confirmed via Galbraith's book.

R6.10 literature closure: status of 47 citations and cited claims (rounds 4-5),
checked 2026-10-18 against fetched sources (raw/fetch_log.json); full notes in RESULTS.md

confirmed: 13 | confirmed-secondary: 4 | bib-only: 14 | corrected: 8 | mismatch: 7 | unverified: 1 | withdrawn: 0

id	group	citation / claim	status	what the fetched source says vs. our use
L1	4E test	US 7,257,709 B2 / US 2004/0114760 A1 (Brown, Yamada; ...	mismatch	h=4 test is Tr(x)=0, x!=0, Tr(x*H(b/x^2))=0: already sqrt...
L2	4E test	King, PKC 2004, 333-345	bib-only	compression paper; cofactor-4 halving test (its Sec. 5 pe...
L3	4E test	Eagle-Galbraith-Ong, ePrint 2009/086	corrected	journal version Adv. Math. Commun. 5(1) (2011) 1-10; cofa...
L4	4E test	Knudsen, ASIACRYPT 1999, 135-149	confirmed-secondary	criterion Tr(x)=Tr(a) stated in FHLM 2004; Tr(a)=0 case d...
L5	4E test	Fong-Hankerson-Lopez-Menezes, IEEE TC 53(8) 2004	confirmed	full text fetched
L6	4E test	'Schroeppel' (no reference given)	corrected	Schroeppel 2000 workshop talk and WO 01/35573 A1, per FHL...
L7	4E test	Pornin, ePrint 2022/1164	confirmed	abstract: halving-based subgroup membership on twisted Ed...
L8	4E test	ePrint 2026/749 'Divide-and-Pair' as a survey	corrected	a method paper (Dai, El Housni, Koshelev, Reijnders), not...
L9	4E test	ECC2K-130 subgroup order r(131)	confirmed	equals l in 'Breaking ECC2K-130' (fetched)
L10	AGM	Fuchsbauer-Kiltz-Loss, CRYPTO 2018	confirmed	abstract: assumptions equivalent to DLog in AGM; Definiti...
L11	AGM	Fuchsbauer-Plouviez-Seurin, EUROCRYPT 2020	bib-only	AGM+ROM confirmed; 'representations for oracle-query elem...
L12	AGM	Bauer-Fuchsbauer-Loss, CRYPTO 2020	bib-only	121-151
L13	AGM	Agrikola-Hofheinz-Kastner, EUROCRYPT 2020 (was recall...	bib-only	96-126
L14	AGM	Zhandry, CRYPTO 2022 (was recalled)	bib-only	66-96
L15	AGM	Duman et al., PKC 2023 (was recalled)	confirmed	abstract: generic group action model for CSIDH-type actio...
L16	AGM	Shoup 1997; Nechaev 1994; Maurer 2005	bib-only	standard; not re-read
L17	class invariance	Jao-Miller-Venkatesan, ASIACRYPT 2005; ePrint 2004/312	confirmed	random reducibility under GRH, (nearly) same End only; eP...
L18	class invariance	Galbraith-Hess-Smart, EUROCRYPT 2002, LNCS 2332, 29-44	confirmed	abstract: isogeny to a GHS-weak curve
L19	class invariance	GHS02 transport cost 'O~(q^1/4)' (16.7 sec. 7.4)	mismatch	O(q^{n/4+eps}) is for two given curves; MMT: finding a we...
L20	class invariance	Gaudry-Hess-Smart, J. Cryptology 15 (2002) 19-46	confirmed	abstract fetched; Thm (via MMT Thm 1): n odd or m=n or Tr...
L21	class invariance	Menezes-Teske-Weng, CT-RSA 2004 (was recalled)	mismatch	abstract: every instance over weak fields is easier than ...
L22	class invariance	'Menezes-Qu condition' n odd or m=n or Tr(a)=0 (R5.2)	corrected	the condition is GHS02's theorem (MMT Thm 1); Menezes-Qu ...
L23	class invariance	'cover genus >= 2^(N-1)-ish' for prime N (R5.4)	mismatch	MMT Thm 11 (from Menezes-Qu): least m>1 is t+1, t = ord_N...
L24	class invariance	Diem 2003, GHS in odd characteristic (was recalled)	corrected	J. Ramanujan Math. Soc. 18(1) (2003) 1-32 (bibtex fetched...
L25	class invariance	Galbraith, LMS JCM 2 (1999) 118-138	confirmed	O(p^1/4 (ln p)^13/2) when conductor is smooth; worst O(p^...
L26	class invariance	Teske J.Crypt 19; Kohel thesis 1996; Miret et al. AMC...	bib-only	bibliographic data fetched; Kohel title page read
L27	image / symm.	Faugere-Huot-Joux-Renault-Vitse, EUROCRYPT 2014	confirmed	PRIOR WORK: Sec. 3.2 has the m^{n-1} decomposition loss a...
L28	image / symm.	Faugere-Gaudry-Huot-Renault, J. Cryptology 27 (2014) ...	bib-only	2-torsion symmetry in Edwards/Jacobi models, as credited ...
L29	image / symm.	Galbraith-Gebregiyorgis, INDOCRYPT 2014	confirmed	binary Edwards; rho still faster for prime degree; '4-tor...
L30	image / symm.	Huang-Petit-Shinohara-Takagi, IWSEC 2013 / PJMI 2015	mismatch	titles differ by venue; abstract: FPPR, prime degree, spe...
L31	image / symm.	Semaev ePrint 2004/031; Gaudry JSC 2009; Diem Composi...	bib-only	bibliographic data fetched; Semaev abstract read
L32	multi-target	Kuhn-Struik, SAC 2001, 212-229: sqrt(2Lr)	confirmed-secondary	Galbraith's book: sqrt(2rL) for 1 < L < r^1/4; holds at L...
L33	multi-target	Fouque-Joux-Mavromati, ASIACRYPT 2014	bib-only	420-438
L34	multi-target	Bernstein-Lange, INDOCRYPT 2012	confirmed	T = l^1/3 table, 1.77 l^1/3 per DLP
L35	sqrt-elu	Bernstein-De Feo-Leroux-Smith, ANTS XIV 2020	corrected	add venue: Open Book Series 4 (2020) 39-55; O~(sqrt l) co...
L36	sqrt-elu	Velu, C. R. Acad. Sci. Paris 273 (1971)	unverified	not fetched; recalled
L37	genus 2 / costs	Gaudry, EUROCRYPT 2000, 19-34	confirmed	O~(q^2), g!q relation term (book)
L38	genus 2 / costs	'Gaudry g!*q^(2-2/g)' (R5.2 model M2)	mismatch	g! is Gaudry-2000's relation term; q^(2-2/g) is GTTD; pro...
L39	genus 2 / costs	sec 13.4 'genus 2 costs O~(q) (Gaudry)'	corrected	O~(q) at g=2 is GTTD double large prime; Gaudry 2000 alon...
L40	genus 2 / costs	Gaudry-Thome-Theriault-Diem, Math. Comp. 76 (2007)	confirmed-secondary	O~(q^(2-2/g)) per Galbraith's book
L41	genus 2 / costs	Enge-Gaudry L[1/2, sqrt2], Acta Arith. 102 (2002)	confirmed-secondary	constant in Galbraith's book and Chi-Dominguez et al. 202...
L42	genus 2 / costs	Nagao, ANTS IX 2010, 285-300	bib-only	extension-field factor base x in F_q (search summary only)
L43	membership	Koshelev, J. Cryptogr. Eng. 2023	corrected	13(1) 125-128 (online 2022) plus correction 14 (2024) 127...
L44	other sec 13-15	Jacobson-Koblitz-Silverman-Stein-Teske, DCC 20 (2000)	mismatch	obstruction is a bound on relation coefficients (from can...
L45a	other sec 13-15	Duursma-Gaudry-Morain ASIACRYPT 1999; Galbraith-Gaudr...	bib-only	bibliographic data fetched
L45b	other sec 13-15	Castryck-Decru EUROCRYPT 2023; Kani J. reine angew. 4...	bib-only	added to section 18 (cited in 13.4)
L45c	other sec 13-15	Smart J.Crypt 1999; MOV IEEE IT 1993; Frey-Ruck Math....	bib-only	added to section 18 (cited in 13.1-13.2)

Figure 1: Status of every checked citation and cited claim (rows L1–L45c; see `RESULTS.md` for each entry). Generated by `plot_claims_status.py` from `claims_status.json`. "bib-only" means the bibliographic data was fetched but our use of the source is unverified.

Prior work for our main results

Claim	Finding (queries in `RESULTS.md`)
Oracle-dependent kernel-saturation penalty	Anticipated: FHJRV EUROCRYPT 2014 §3.2. Earlier 2-torsion symmetry work: Faugère–Gaudry–Huot–Renault, J. Cryptology 2014. Our exact census and measured class factors are what remains ours.

Trace pair as an E/4E coset label	Nothing found. Nearest: the x-only h = 4 test of US 7,257,709. King §5 and the EGO text were not read.
Isogeny-augmented generic group model	Nothing found for isogeny oracles. Related: Duman et al. PKC 2023 (group actions); Rupp et al. ASIACRYPT 2008 (extended generic operations; coverage unverified).
AGM with point-evaluation isogeny oracles	Nothing found. arXiv 2608.01675, which a search summary suggested, defines no such model.
Class-wide multi-target rho via transport	Nothing found as a stated result. The ingredients are Kuhn–Struik and Galbraith 1999.

"Nothing found" means the listed queries returned nothing relevant. It does not show that no such work exists.

Limits and next step

Primary texts were not read for King, EGO, Knudsen, Kuhn–Struik, Nagao or Menezes–Qu, because the fetches were blocked. The entries rest on abstracts or secondary sources, as stated. No claim about any deployed curve is made.

Next: apply `references_proposed.md` to §18, and apply the wording fixes listed in `RESULTS.md` § "Next step".